

Vision-Based Gaze Estimation: A Review

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Abstract—Eye gaze is an important natural behavior in social interaction as it delivers complex exchanges between observer and observed, by building up the geometric constraints and relation of the exchanges. These interperson exchanges can be modeled based on gaze direction estimated using computer vision. Despite significant progresses in vision-based gaze estimation in last 10 years, it is still nontrivial since the accuracy of gaze estimation is significantly affected by such intrinsic factors as head pose variance, individual bias between optical axis and visual axis, eye blink, occlusion and image blur, degrade gaze features, lead to inaccurate gaze-involved human social interaction analysis. This article aims to review and discuss existing methods addressing above-mentioned problems, gaze involved applications and data sets against the state of the arts in vision-based gaze estimation. It also points out future research directions and challenges of gaze estimation in terms of metalearning, causal inference, disentangled representation, and social gaze behavior for unconstrained gaze estimation.

Index Terms—3-D gaze estimation, computer vision, head pose, optical axis, visual axis.

I. INTRODUCTION

GAZE plays an important role in human behaviors as it is capable of perceiving ambient objects and human intentions. By estimating gaze direction, human visual attention and cognitive process can be obtained and analyzed. Hence, gaze estimation is meaningful for human-centered science and technology. In the field of computer vision and human-computer interaction (HCI), studies have tried model-based or appearance-based methods with artificial features to learn gaze direction from RGB images. With the development of deep learning technology like convolutional neural network (CNN) and generative adversarial network (GAN), the accuracy and generalization ability of vision-based gaze estimation

was further improved. However, many problems which lower the accuracy of gaze estimation still exist.

Intrinsic problems of gaze estimation are illustrated in Fig. 1. There are mainly three steps for a simple and straightforward approach for gaze estimation under constrained conditions. First, given a full face RGB image, eye regions should be localized by eye detection or face detection. Next, geometry or appearance features are extracted from the detected eye region. Finally, the relationship between extracted gaze features and 3-D gaze directions or gaze points is learned using regression function. Gaze estimation would be accurate in an extremely constrained condition, such as clear backgrounds and limited head movements. However, human natural behaviors and person-specific parameters, such as head pose variance, eye blinking, occlusion, image blur and the offset between optical axis and visual axis, make the simple gaze estimation sensitive to these outliers. Specifically, the above-mentioned five challenges not only affect the robustness of eye region detection, but also add noises to the gaze feature extraction. Therefore, the regressor cannot be well built from extracted nonrepresentative gaze features. Furthermore, gaze-based human-human social interaction and cognition reasoning is influenced.

Detailed review work have been shared in the past 10 years. Dan and Qiang [1] presented a detailed review about gaze estimation including eye detection modules and artificial gaze features. Kar and Corcoran [2] compared various gaze estimation user platforms, algorithms as well as performance evaluation methods and propose a standard framework for the evaluation of different gaze tracking systems. Cazzato *et al.* [3] reviewed the history of eye-tracking technology and improvements of cutting-edge computer vision advancements on gaze estimation in first-person view and third-person view. Akinyelu and Blignaut [4] focus on appearance-based gaze estimation using deep learning algorithms and analyze different classic CNNs. Despite previous detail and thorough reviews, an analysis of intrinsic problems of gaze estimation is significant to beginners in this field. The main aim of this review is to survey the existing challenges of gaze estimation and corresponding solutions, and to compare existing data sets, applications and possible research directions. Compared to previous reviews, our main contributions are as follows.

- 1) An updating review of gaze estimation in terms of existing key challenges and corresponding solutions, data sets and applications.
- 2) A prediction of future possible research directions of gaze estimation.

The following of this article is organized as follows. Section II reviews main challenges, existing corresponding

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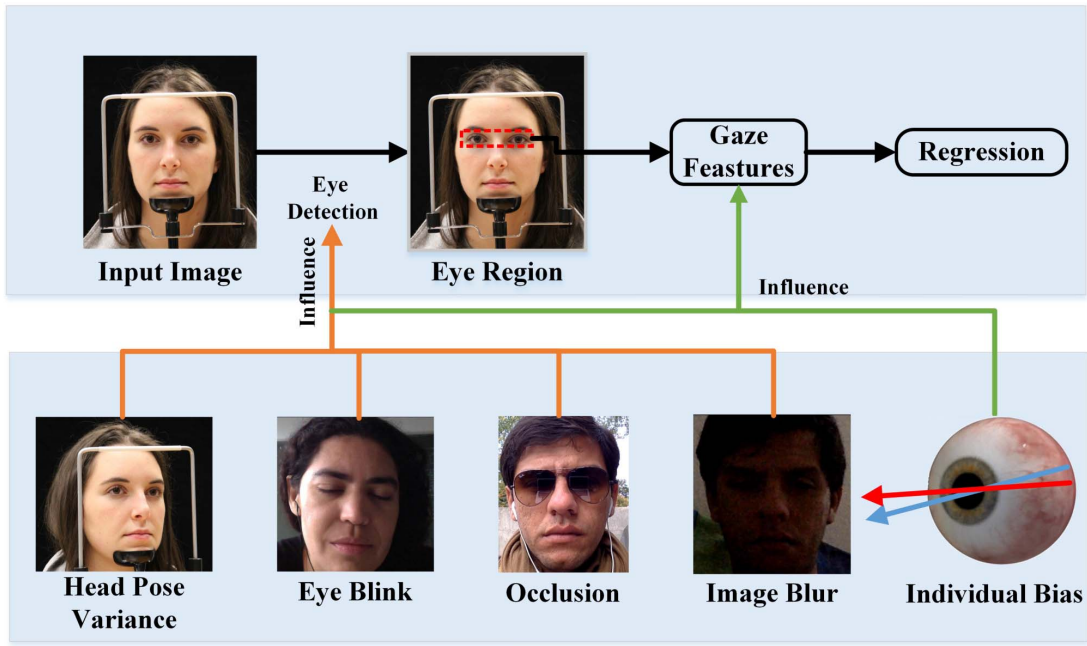


Fig. 1. Overview of a simple gaze estimation which consist of eye detection, feature extraction, and gaze regression. Challenges of gaze estimation are head pose variance, eye blink, occlusion, blur images and individual bias.

solutions and some novel observations and ideas in gaze estimation. Section III summarizes recently released data sets for 3-D gaze estimation in terms of gaze, head pose and data collection configurations. In Section IV, applications of gaze estimation in HCI are surveyed. In Section V, we discuss possible research directions in future based on advanced progresses in computer vision. Finally, the conclusion of this review is given in Section VI.

II. CHALLENGES AND SOLUTIONS

Problems showed in Fig. 1 affect the performance of existing gaze estimation methods, including head pose variance, individual bias, eye blink, occlusion and image blur. This section analyzes the influences of these problems and current solutions in literature.

A. Head Pose Variance

It is very important to consider head pose when estimating human gaze, since 70% of visual gaze is contributed by head pose [31]. As a component of head, both eyes rotate and translate along with head in world coordinate system (WCS). Therefore, head pose can determine gaze direction coarsely. However, head pose is not enough to describe human gaze since eye can rotates. When gazing at the target in front while moving their head, viewers need to rotate their eyes, causing angular offsets between eye gaze and head pose. An observation is that the offset between eye gaze and head pose is larger when human gaze at objects on the right than when they gaze at objects in the front [32], [33]. Apart from considering head pose, eye gaze related features need to extracted from eye regions to describe human gaze finely. Thus, robust gaze estimation need to extract representative information from head pose and gaze to describe human gaze multigranularly.



Fig. 2. Eye appearance variations with the same gaze angle under different head poses (yaw: 30°, 0°, -30°) generated using UnityEye. For example, from left to right, the relative position between iris center and eye corners change significantly.

One affect of head pose variance is that the change of eye position. To extract highly relevant gaze features, eye regions or face regions should be cropped from original images to subtract noisy backgrounds. Face landmarks are usually treated as reference points to crop the desired area. Then, gaze features can be extracted but still suffer from appearance changes.

Another impact of head movements is that eye appearance changes significantly when head movements are considered. Ranjan *et al.* [7] illustrated this change in detail, which is shown in Fig. 2. Subjects' eye appearance differs significantly when gazing at a same target under different head poses. For example, the iris region tends to be complete when the head pose near 0° while part of the iris region is occluded when the head pose near -30° and 30°. This appearance variance would result in nonrepresentative gaze features. The distributions of gaze and head pose in existing data sets are nonuniformly distributed, which also lead to overfitting between gaze and head pose. Supposing that pose insensitive features are not extracted, the trained model would failed to estimate gaze when training and testing samples disobey i.i.d (independent identically distribution) hypothesis.

Many studies are proposed to mitigate the influence of head pose on gaze, and they are reviewed from Sections II-A1–II-A5.

1) *Cluster-Based Method*: One approach against the head pose variance is to cluster head poses and train multiple regression models for each cluster.

Sugano *et al.* [5] proposed to reconstruct eye regions from multiple cameras and then train a random regression forest model to learn the mapping function between eye region and eye gaze. Specifically, each regression tree in regression forest model is trained for each redundant cluster. The proposed algorithm achieves the mean error of 6.5° on the UT-Multiview data set. Similar cluster scheme is also used in [6] to tackle the head pose variance. However, training multiple models for each cluster would increase the complexity of computation. Inspired by the cluster scheme, a novel approach is proposed in [7], which is a CNN model mainly composed of five convolution layers (conv1–conv5) and one fully connected layer (fc1). Different from training multiple regression models, this CNN model updates weights of conv1–conv5 and fc1 according to gradients to learn gaze representation, and then multiple CNN streams are further updated after fc1 using the learned gaze representation and head pose vectors from corresponding head pose clusters. Each stream has two fully connected layers after fc1. This scheme makes full use of limited data to learn a basic gaze representation, followed by adjusting the representation according to different head poses. It achieves the mean error of 5.65° on the Columbia data set and 5.3° on the MPIIGaze data set.

2) *Face Frontalization-Based Method*: The main idea of face frontalization-based methods is to reconstruct a 3-D face model of human face. The face can be transformed to be frontal so that the head variance can be eliminated.

Shi *et al.* [8] proposed a symmetry prior knowledge-based 3-D reconstruction for gaze estimation. Given 3-D mesh data and videos from multiple cameras, this method first detects a face region in videos and 3-D mesh data. And then, it extracts key points of a face in order to obtain a symmetry plane. Second, based on the symmetry plane, a 3-D face model is reconstructed using symmetry prior knowledge and then a virtual frontal face image is rendered using super solution rendering. Finally, based on the assumption that the diameter of iris equals to the radius of the eyeball, a 3-D iris position and a 3-D eyeball center are obtained by mapping a 2-D iris center to a 3-D eyeball model. Thus, the gaze direction is the line connected from the 3-D eyeball center to the 3-D iris center. Similarly, Funes-Mora and Odobez [9], [10] proposed to first learn a person-specific 3-D mesh model based on 3-D morphable model in the offline stage, and then in the online stage, head pose is estimated from depth data. The frontal face can be further generated using head pose and 3-D mesh data. Finally, gaze can be learned from the cropped eye images and then mapped back to WCS. Ghiass and Arandjelovic [11] proposed an accurate head pose estimation. A person-specific morphable model is used to estimate head pose in depth space using iterative closest point algorithm. Then, synthetic frontal faces are obtained by transformation and rerendering. Finally, gaze directions are regressed from appearance features in synthetic

images. The proposed method achieves the mean error of 8.9° while subjects move their head. In [12], dense markers on face are detected and the cascade regression is used to bridge 2-D to 3-D. Then, 3-D faces are reconstructed by solving an optimization problem. After face frontalization, binary features are extracted from six eye counter markers and one pupil marker in order to train a linear support vector regressor, mapping gaze features to gaze directions. It achieves the mean error of 4.2867° on the UT-multiview data set. Although the face frontalization method can generate a frontal face, if part of an eye region is not captured, the eye appearance is still incomplete, which degrade gaze estimation.

3) *Feature Fusion-Based Method*: Feature fusion can improve the performance of classification or regression by improving feature representation ability via considering multimodel features. Therefore, the fusion of head pose vectors and gaze features is also an effective way to against the head pose variance.

Jafari and Ziou [13], [14] used a kinect to estimate head pose and an active HD camera to capture eye regions. Then, iris displacement vectors and reference points are extracted from the eye region. Finally, using the combination of iris displacement vectors, head positions and head poses, gaze directions are estimated using variational Bayesian multinomial logistic regression. The progress of deep learning attracts appearance-based gaze estimation in recent years. Zhang *et al.* [15] proposed to use CNN to extract gaze representation from eye images. Specifically, it first crops an eye region by rotating and translating the camera to the eye center. Then, normalized eye images and head pose vectors are fed to a Lenet model, to learn the regression from gaze representation along with head pose vectors. This approach achieves the mean error of 6.3° on the proposed MPIIGaze data set. The performance is further improved to the mean error of 5.5° by [16] using a VGG-16 model replacing the Lenet model. In the MPIIGaze data set, right eyes are horizontally flipped. However, Lemley *et al.* [18] proved that the performance can be improved by treating both eyes as ground truth instead of flipping one to the other. Apart from that, by replacing 5×5 convolutions with 3×3 convolutions and using the relu activation as well as fusing eye images and head pose vectors as the input of CNN, the mean error of gaze estimation is further improved to 3.65° on the MPIIGaze data set. Sun *et al.* [17] proposed a multistream CNN model, which consist of four parallel VGG models extracting gaze features and head poses from both eye and face regions. The final gaze output is the average of all streams. The result is the mean error of 6° on the UT-multiview data set. Lian *et al.* [19] proposed a multistream multitask CNN model-based gaze estimation. The proposed model learn gaze features from both cropped eye images and head pose information from full face in RGB images and depth maps refined by GAN. The learned head pose information is then concatenated with gaze features. It achieves the mean error of 4.8° and 38.7 mm on the EYEDIAP and the ShanghaiTechGaze data set, respectively. Above studies employ the scheme of simply fusing gaze features and head poses in a serial way, which is simple and straightforward.

4) *Geometric Relationship-Based Method*: This section reviews researches using geometric knowledge of eyeball models and transformation between different coordinate systems to compensate deviation caused by head movements.

Human head and eye ball both have six degrees of freedom (DOF), sharing four DOFs which are translation along x -axis, y -axis, z -axis, and rotation along z -axis. The normalization method proposed in [5] rotates and translates a camera so that the camera looks at an eye center and its x -axis is parallel to the x -axis of head coordinate system (HCS), leading to only two DOFs in the normalized space. In this normalized space, gaze direction can be represented using pitch and yaw, while roll is always zero. By adjusting the focal length, normalized images would have a same size. This normalization method converts images from 6D space to 2-D space using perspective warp transformation, which mitigate appearance variations caused by head movements. Most existing state of the arts employ this normalization or the improved version [20] before input images to neural networks. Zhou *et al.* [21] proposed a two-eye model-based gaze estimation method where eye centers can be inferred from a 3-D human face model. Using the prior knowledge that the relative position of 3-D eye ball center is fixed, once personal parameters are calibrated and the face is detected, the eye center can be estimated, followed by calculating the optical axis and the visual axis. Due to small user-head movements, the estimated iris center could changes. Zhang *et al.* [22] proposed to use a back propagation (BP) neural network to rectify the estimated iris center for gaze estimation. Specifically, the proposed method first detects an eye region using Haar-like feature template and then localize the iris region and the pupil region using rectangular template with threshold. The iris contour is segmented using horizontal template edge detection and RANSAC, in order to infer gaze directions from the minor axis of iris contour, the iris center and the 3-D eyeball model. The BP network is then used to correct the estimated iris center using rotation and translation vectors which obtained from two images in sequence. Xiong *et al.* [27] rectify 3-D Pupil-Eye-Contours-based features using head rotation and translation vectors. The proposed method first estimates a frontal face from multiple images using active shape models. Then, the rotation and translation are obtained by optimizing the error between a test face and the estimated frontal face. Finally, 3-D Pupil-Eye-Contours-based features are extracted from the rectified face structure for estimating the gaze direction using adaptive linear regression. Lu *et al.* [23], [24] proposed a decomposition method to mitigate head rotation. The main idea of this approach is to estimate an initial gaze direction under a fixed head pose, then the initial head pose is rotated to a predefined head pose, so that the initial gaze direction can be transformed to another under the defined head pose. Thus, the true gaze direction in WCS under the predefined head pose can be compensated by calculating the angle between the initial head pose and the defined head pose. Zhang *et al.* [25] proposed a gaze estimation method (GEMHSSO) that can adapt to natural head movements with a single camera and a single light source. When subjects move their head from a calibration point, according to the relationship between interpupillary distance

and pupil-glint vectors derived from imaging principle and geometric relation, the pupil-glint vector under moved head poses is automatically translated to the pupil-glint vector at the calibration position. Thus, gaze estimation errors caused by head movements are mostly eliminated. The HCS created by additional face detection modules could introduce errors to gaze estimation when gaze is transformed from HCS to camera coordinate system (CCS), since these modules are usually not robust when subjects have emotions [26]. Deng and Zhu [26] proposed to learn head poses and gaze directions from full faces and eye regions, respectively. The novelty of this work is that a nonlearnable gaze transform layer which converts gaze from HCS to CCS.

5) *Autoencoding-Based Method*: Additional face detection modules could introduce errors to gaze estimation when estimating head pose vectors. An alternative approach is to encode head pose information along with gaze information by learning gaze related representation automatically.

Zhang *et al.* [28] proposed a novel CNN model which learns the gaze related representation directly. The spatial attention module in the proposed CNN model weights eye regions for eliminating irrelevant area noises. Meanwhile, the head pose information is also encoded since it hides in the full face. The proposed approach achieves the mean error of 4.8° and 6° on the MPIIFACEGaze data set and the EYEDIAP data set, respectively. Wang *et al.* [29] proposed an adversarial learning-based model for learning the head pose and appearance-insensitive gaze representation. The main idea of this work is to feed gaze features learned from CNN to a gaze estimator, an appearance classifier and a head pose classifier, aiming at confusing the appearance classifier and the head pose classifier while keeping the gaze estimator a good performance. This approach achieves the mean error of about 4.3° , 5.4° , and 9.9° on the MPIIGaze, the UT-multiview and the EYEDIAP data set, respectively. Wang *et al.* [30] proposed a multitask CNN model which detect head poses and gaze directions simultaneously. This end-to-end model first extracts face and eye regions using a multistream CNN, and then head poses and gaze directions are learned in a parallel architecture. Head pose hidden features are added to gaze hidden features in fusion layers, which provide the gaze representation with the head movement information. The proposed approach achieves the mean error of 0.82° , 4.02° on the ColumbiaGaze, the MPIIGaze, respectively.

Overall, there are mainly five methods addressing the head pose variance in gaze estimation tasks, as shown in Table I containing used methods, references, input data, evaluated data sets and results. Cluster-based methods aim to train multiple models for each head pose cluster, which encode relationship between gaze features and gaze directions under specific head poses. The advantage of this method is that the cluster scheme is easy to be implemented. Most of face frontalization methods need depth data to generate a 3-D face model or align the face model using dense points. Then, transformation from the original head pose to the frontal head pose is conducted. But the unseen part of a face due to self-occlusion still cannot be recovered, thus causing missing appearances. Feature fusion is a common way in machine learning to improve accuracy.

TABLE I
CLASSIFICATION OF 3-D GAZE ESTIMATION METHODS ADDRESSING HEAD POSE VARIANCE

Problems	Method category	References	Input			Dataset	Accuracy
			Eye	Face	Other		
Head pose variance	Cluster-Based Method	[5]	✓		head pose vector	UT-Multiview	6.5°
		[6]	✓		head pose vector	unknown	2.9°
		[7]	✓		head pose vector	ColumbiaGaze MPIIGaze	5.65° 5.3°
	Face Frontalization -Based Method	[8]		✓	multiview video	unknown	yaw:0.3266(rag) pitch:0.4541(rag)
		[9]	✓	✓	head pose parameters	unknown	average 26.2°
		[10]	✓	✓	head pose parameters	EYEDIAP	FT-PS:7.05°CS-PS:2.75°
		[11]	✓	✓	head pose parameters	EYEDIAP	8.9°
		[12]	✓		head pose	UT-Multiview	4.2867°
		[13]	✓	✓	head pose parameters	unknown	96%
	Feature Fusion -Based Method	[14]	✓	✓	head pose parameters	unknown	95.83%
		[15]	✓		head pose vector	UT-Multiview MPIIGaze	5.9° 6.3°
		[16]	✓		head pose vector	MPIIGaze	5.5°
		[17]	✓		head pose vector	UT-Multiview	6°
		[18]	✓		head pose vector	MPIIGaze	3.65°
		[19]	✓	✓	depth	ShanghaiTechGaze EYEDIAP	38.7mm 4.8°
	Geometric Relationship -Based Method	[5]	✓		head pose vector	UT-Multiview	6.5°
		[20]	✓		head pose vector	MPIIGaze	6.6°
		[21]		✓		unknown	1.99°
		[22]	✓			unknown	0.8°
		[23]	✓		head pose parameters	unknown	2.38°
		[24]	✓		head pose parameters	unknown	2°-3°
		[25]	✓		one light source	unknown	1.5°
		[26]	✓	✓		unknown	4.3°
	Auto-Encoding -Based Method	[27]	✓	✓		unknown	3.3°
		[28]		✓		MPIIFACEGaze EYEDIAP	4.8° 6°
		[29]	✓			MPIIGaze EYEDIAP	4.3° 9.9°
						UT-Multiview	5.4°
		[30]	✓	✓		ColumbiaGaze MPIIFACEGaze	0.82° 4.02°

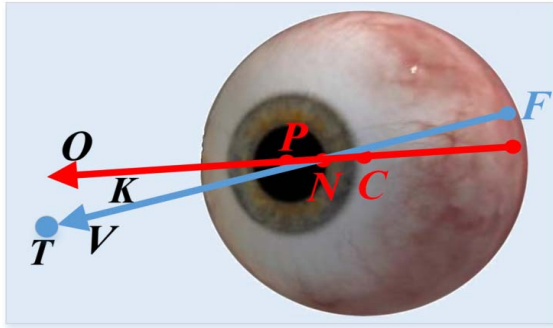


Fig. 3. Offset K exists between optical axis and visual axis in a 3-D eyeball model. A general eye gaze estimation method always has estimation deviations if this individual bias is not considered.

The model can learn the mapping between eye appearances and gaze directions from the correspondence between head poses and gaze-related features. Meanwhile, head pose and gaze can also be overfitted due to the domain-specific training process. Geometric relationship-based methods normalize input images mostly using perspective warp transformation for limiting head poses or use geometric restrictions across coordinate systems to compensate gaze direction deviation caused by head movements. Different from using head pose

vectors, autoencoding-based methods extract head pose representation in hidden layers along with gaze representation. In future, to tackle the overfitting of head pose and gaze, these two factors can be distangled by learning the redirection relationship between head pose and gaze from existing domains. Though available data is limited, the gaze direction under a unseen head pose can be modeled using the learned redirection mapping.

B. Person-Specific Bias

Funes-Mora and Odobez [38] illustrated this person-specific bias in detail, as shown in Fig. 3. When people gaze a target T in 3-D space, the eyeball rotates around the eyeball center C until the visual axis (V) intersects the target T . Visual axis is the line connecting the target T and the fovea F (locating on the retina). It intersects the optical axis (O) at the point N . O is the line connecting C and the pupil center P . According to Funes-Mora's description, $V = O + K$, where K is the person-dependent offset between V and O . This offset is person specific, which generates prediction errors if the designed model do not take it into consideration. Existing methods are reviewed from Sections II-B1–II-B3.

1) *Parameters Estimation-Based Method*: If the offset K can be estimated, gaze estimation can be more accurate.

The person-specific offset K can be inferred from parameters in a 3-D eyeball model. Model and Eizenman propose a gaze estimation method using two cameras and two light sources [34], [35]. First, the optical axis is estimated as the line connecting the center of curvature of the cornea to a virtual pupil center. Under the assumption that at each time instant the visual axes of both eyes intersect on a observation surface, the offset K can be calculated by optimizing the distance between two eyes' gaze points on a screen. Maio *et al.* [36] proposed a constraint-based gaze estimation method without active calibration. Under the proposed constraints, the binocular constraint, the screen constraint and the subject independent parameter range constraint, constrained eye parameter estimation can be achieved by solving an inequality-constrained optimization problem using quadratic programming. Reale *et al.* [37] proposed a 3-D eyeball model parameter estimation method. The proposed method has two calibration steps. In the first step, two iris positions are obtained by asking subjects to gaze the camera and a calibration target. Then these iris points are converted into two normalized vectors, pointing from the camera center to the world image plane. The fovea point is estimated in the second calibration step by intersecting the vector pointing from the camera center to the 3-D eyeball center with a sphere centered at our estimated eyeball center. After the calibration step, the optical axis can be calculated by solving a linear equation of 3-D iris contour vectors and an approximation of optical gaze vectors. The visual axis is then obtained by connecting the fovea point and a new 3-D iris center which is estimated by intersecting the optical axis with the eyeball sphere. In [38], the optical axis is estimated through an appearance generative process of eye images, and the visual axis is then inferred by fixed eye parameters and variational Bayes. Lindén *et al.* [40] proposed to learn person-specific calibration parameters along with gaze representation. At the training stage, the proposed CNN model take both eyes, a full face and calibration parameters as input. At the testing stage, for an unseen sample, only a few calibration samples ($< nine$) are needed to learn calibration parameters. With nine calibration samples, the proposed method achieves the mean error of 2.7° on the MPIIGaze data set. Xiong *et al.* [39] decomposed the gaze representation into fixed effects and random effects. By solving the nonlinear mixed effects model, this approach can estimate the offset K . Giving 200 calibration samples, it achieves the mean error of 3.8° on the MPIIGaze. Other than the offset between optical axis and visual axis, Chen and Shi [41], [42] analyzed that there exists bias that can not be learned from images. The true gaze is decomposed into the visual axis and the other person-specific bias. If no calibration images are available, this bias is set to zero. With calibration, the bias is the average of all calibration samples' difference between the visual axis and the true gaze. Using dilated convolutions and the decomposition method in [41], it achieves the mean error of 3° with nine images of a single calibration target. Unlike other estimation methods, the offset K can also be set to a constant according to statistics, which is 5° horizontally and 1.5° vertically [43].

2) *Gaze Redirection-Based Method*: The main idea of gaze redirection-based method is to learn gaze changes of one

person, and then the person-specific bias can be calibrated using these redirected samples.

Liu *et al.* [44] proposed a novel method which estimates the angle difference between two eye images with different gaze directions from a same subject. At the training stage, two images with ground truth are fed into a CNN with two parallel streams to learn gaze representation. Then, hidden features of both images are concatenated before outputting a differential angle. The two input images must be collected from a same person so that the accurate differential angle can be estimated by subtracting one gaze representation to another, without the impact of individual bias changes. At test stage, the differential angle between a calibration sample and a test sample is calculated, then the gaze direction of the test sample is the sum of the estimated differential angle and the ground truth of the calibration sample. The proposed method achieves the mean error of 3° , 3.8° , and 3.77° on the EYEDIAP, the MPIIGaze and the UT-multiview data set, respectively. The differential information can also be learned to redirect gaze directions, generating more calibration samples. Yu *et al.* [45] proposed a gaze redirection network for generating more calibration samples with minimum person-specific bias changes. The proposed network takes a source image and the differential angle between the source image and a target image as inputs. By learning vertical flows and horizontal flows, the network outputs learned redirected gaze images. The proposed model is trained on synthetic samples generated by the UnityEye. With nine calibration samples, the trained model achieves the mean error of 2.6° and 4.01° on the ColumbiaGaze and the MPIIGaze data set. Yu and Odobez [46] further propose the first unsupervised gaze estimation network using gaze redirection. Park *et al.* [47] also proposed a person-specific gaze estimation based on gaze redirection. The encoder-decoder model rotates the gaze and head pose representation learned from encoder using rotation matrices, and the loss between the decoded approximation and the ground truth is calculated to train the whole model. To alleviate the overfitting problem caused by simply fine-tuning a trained model with available calibration samples, a person-specific model is obtained using the metalearning method MAML. With nine calibration samples, the proposed method achieves 3.14° on the MPIIGaze data set.

3) *Gaze Pattern-Based Method*: Gaze patterns provide potential gaze points when subjects view visual stimulates in images, which can be used to calibration.

Chen and Ji [48] proposed a probabilistic 3-D gaze estimation method based on saliency maps and a 3-D eye model. Unlike traditional explicit calibration approaches, the proposed approach estimates the probability distribution of eye parameters and eye gaze. The saliency map of a given image is considered as the conditional probability of gaze positions. Then, using a Bayesian network, the probability distribution of the offset K is estimated from a saliency map, followed by estimating the likelihood gaze map. Finally, gaze points are selected as the maximum posterior points on the product of the saliency map and the likelihood gaze map. Based on the observation that gaze patterns of individuals are similar for a certain stimulus, Alnajjar *et al.* [49], [50] proposed a gaze

TABLE II
CLASSIFICATION OF 3-D GAZE ESTIMATION METHODS ADDRESSING INDIVIDUAL BIAS

Problems	Method category	References	Calibration		Dataset	Accuracy
			Active	Passive		
Individual bias	Parameters Estimation Based Method	[34]		✓	unknown	1.3°
		[35]		✓	unknown	0.5°
		[36]	nine-point calibration		unknown	L:2.825° R:3.075°
		[37]	20 gaze markers		unknown	1.0889°
		[38]		✓	unknown	fixed:3.4° moving:6.3°
		[39]	200 calibration samples		MPIIGaze	3.8°
		[40]	nine calibration samples		MPIIGaze	2.7°
		[41]	nine calibration samples		MPIIGaze	3°
		[42]	nine calibration samples		MPIIGaze	2.7°
		[43]	✓		unknown	<10°
	Gaze Redirection Based Method	[44]	nine reference samples		EYEDIAP MPIIGaze UT-Multiview	3° 3.8° 3.77°
		[45]	nine calibration samples		MPIIGaze ColumbiaGaze	2.6° 4.01°
		[46]	100 calibration samples		EYEDIAP ColumbiaGaze UT-Multiview	7.09° 5.25° 6.8°
		[47]	nine calibration samples		MPIIGaze	3.14°
	Gaze Pattern Based Method	[48]		✓	unknown	1.77°
		[49]		✓	unknown	4.2°
		[50]		✓	unknown	4.3°
		[51]		✓	unknown	2.57°

pattern-based auto calibration method. Initial gaze points are estimated using a offline trained 2-D manifold. Calibration is achieved by optimizing the loss between new gaze points and its k closest neighbors. In [51], the uncalibrated gaze pattern is recovered using eye alignment, manifold construction and gaze pattern regularization.

In summary, the parameter estimation-based method, the gaze redirection-based method and the gaze pattern-based method are three approaches to improve gaze estimation considering the person-specific bias, as shown in Table II including used methods, references, calibration methods, evaluated data sets and the accuracy. The individual bias exists in human eyeball including the offset between the optical axis and the visual axis. Parameter estimation-based methods aim to learn the mapping between extracted features and the offset K in the calibration process using geometric knowledge or fine-tuning the model with calibration samples. Model-based gaze estimation methods mostly depend on calibration points cause the offset K and other eyeball parameters like the eyeball center need to be estimated. In comparison, appearance-based gaze estimation methods mainly depend on calibration images with ground truth. Gaze redirection-based methods learn gaze changes from images of a same person, which encode individual bias in the learned gaze changes. Thus, the trained gaze estimator can adapt to unseen subjects' samples using fine-tuning with their redirected samples. Gaze pattern-based methods use the prior knowledge of visual saliency when subjects are viewing visual stimulates. It is



Fig. 4. Examples of blinking, occlusion, and image blur. These special cases should be taken into account, especially in unrestricted gaze estimation.

based on the assumption that the visual interests of viewers are similar. However, for autism spectrum disorder people whose visual interests are irregular, this assumption may not work. Although this offset K is hard to be learned from images, a gaze estimator without any calibrations is still desired in future.

C. Eye Blinking, Glasses and Blur Issues

Examples of eye blink, occlusion and image blur are shown in Fig. 4. Occlusion and blur images are common problems in computer vision, which usually degrade the estimation accuracy since these outliers can not offer useful gaze-related information. To address these issues, on the data level, GAN can be used to generate synthetic images without wearing glasses or refine blur images. Using multiview data is also an approach to avoid self-occlusion. In feature level, extracting

anti-light/blur/occlusion features can be effective as well. Apart from these problems, eye blinking should also be tackled cause no iris regions can be detected in samples with eye blinking.

For lighting and blur issues, Ge [52] proposed a lighting variance insensitive features, named Gabor directional binary pattern (DBP). Multiscale and multiorientation Gabor wavelets are used to decompose an eye image, followed by the DBP operator, which encode directional differential patterns in 3×3 pixels. Sun *et al.* [53] proposed a pupil center localization method on blur images. Based on the observation that pupil region is darker than other regions, the proposed method uses a pupil template with sliding windows to detect the pupil center. It works well on blur images captured by Kinect2.

Multiview data and GAN are two ways addressing occlusions. Rangesh *et al.* [54] used CycleGAN to eliminate the occlusion and reflection caused by the glasses in driving situation. During the data collection in [55], subjects are asked to wear a pair of eye-tracking glasses to get more precise gaze direction ground truth. Fischer *et al.* [55] proposed to use GAN to recover the eye region illuminated by eyeglasses. Lian *et al.* [56] propose a CNN model which learns 3-D gaze estimation and gaze point estimation simultaneously using multitask learning. The model encodes gaze representation from three views which can alleviate self-occlusion.

Eye blinking should also be detected since iris regions are entirely occluded. Lu *et al.* [57] observed that eye blinkings do not change the eye orientation, but produce unseen blinking eye features, which must exist outside the learned manifold, meaning that weights in interpolated features cannot be optimized to the optimal value. Therefore, eye blinking can be detected by analyzing whether the weight in interpolated features is close to one. Cortacero *et al.* [58] simply used a CNN model to detect eye blinking samples and propose an eye blinking detection data set. Chen *et al.* [59] proposed an unsupervised outliers detection method for detecting eye blinking, occlusion and blur samples in existing data sets, by generating pseudo-labels to adjust the hyperparameter weighting the loss function. The proposed method achieves the mean error of 4.5° on the MPIIGaze data set and 4.1° even though 300 outlier images per person are added to the MPIIGaze.

D. Issue of Insufficient Training Samples

The issue of insufficient training samples still limits the robustness and generalization of existing methods. Synthetic eye images seem to be an alternative data source. Wang *et al.* [60] proposed a hierarchical generative model that enables bidirectional inference. It contains two models, and one is a hierarchical generative shape model (HGSM) which can estimate gaze by learning geometric features and eye shapes and another is a conditional bidirectional GAN (c-BiGAN) which can synthesis eye images. This model achieves the mean error of 7.5° and 8.4° on the MPIIGaze data set and the UT-multiview data set, respectively. Lu *et al.* [61] uses images from four reference cameras and a single-directional flow model to generate eye images with different head poses. Computer graphics software such as the UnityEye can render the eyeball appearances [62]–[64], which can directly generate

eye images and ground truth with varying lighting conditions and head poses. Synthetic redirected eye images can also improve the performance of gaze estimation through generating calibration samples. Details are in Section II-B2.

E. Some Novel Ideas for Gaze Estimation

Gaze estimation can be improved by solving above main problems. However, other novel ideas are also mentioned from Section II-E1–II-E3, which improve gaze estimation by using eye semantic information, two-eye asymmetry and dynamic information.

1) *Landmarks and Semantic Information*: Instead of learning gaze directly from eye images, gaze can be alternatively learned from eye semantic information and eye landmarks. Park *et al.* [65] proposed to learn gaze maps, which are 2-D binary maps of eyeball and iris, from eye images, and then gaze can be regressed from the learned gaze maps. Three hourglass blocks are used to learn gaze maps and five dense blocks are employed to regress gaze direction. This model achieves the mean error of 4.56° , 3.59° , and 10.63° on the MPIIGaze, the ColumbiaGaze and the EYEDIAP data set, respectively. Yu *et al.* [66] proposed a multitask method to learn eye gaze and eye landmarks simultaneously. It first learns the coefficients of the proposed constrained landmark-gaze model, the scale and translation defining the eye region. And then a decoder is used to reconstruct the eye landmark positions and gaze with two sets of parameters and head poses. The proposed method achieves the mean error of 5.7° on the UT-multiview gaze data set and 7.725° on the EYEDIAP data set. By extracting angular features encoding the orientation of pupil centers, Jyoti and Dhall [67] propose to regress gaze from both eye regions and these geometric angular features. It achieves the mean error of 2.22° on the ColumbiaGaze data set. It is easier to learn gaze from landmarks or semantic maps in comparison to complex eye appearances.

2) *Two-Eye Asymmetry*: The observation in [68] indicates that even using a same gaze estimation algorithm, the performance of two eyes is not the same. Hence, Cheng *et al.* [68] proposed to exploit the two-eye asymmetry by weighting one of the eye. Specifically, the asymmetric regression network predicts 3-D gaze directions for both eyes and outputs the loss of both eyes. And the evaluation network adaptively weights the loss of two eyes by evaluating the two eyes according to their performance during optimization. Based on the proposed network, head pose information is further considered in [69] and it achieves the mean error of 4.3° , 5.71° , and 8.4° on the MPIIGaze, the EYEDIAP and the RT-GENE data set. Cheng *et al.* [70] proposed a coarse-to-fine gaze estimation network, which can assign weights for left and right eyes based on extracted face features. The proposed method achieves the mean error of 4.1° and 5.3° on the MPIIGaze and the EYEDIAP data set.

3) *Dynamic Information*: Eye gaze is a natural human behavior containing temporal information which could be useful for improving gaze estimation. Duan *et al.* [71] proposed a gaze estimation method for human object interaction recognition. Head pose information is extracted using Gabor wavelet and principle component analysis, and then multiple frames

TABLE III
CLASSIFICATION OF 3-D GAZE ESTIMATION METHODS ADDRESSING BLINKING, OCCLUSION,
BLUR IMAGES, INSUFFICIENT TRAINING SAMPLES, AND SOME NOVEL IDEALS

Problems	Method	References	Dataset	Accuracy
Blinking, glasses and blur images	Blur Images	[52]	unknown	$<2^\circ$
		[53]	unknown	1.4° - 2.7°
		[54]	unknown	80.49%
	Occlusion	[55]	MPIIGaze	4.3°
			RT-GENE	7.7°
		[56]	ShanghaiTechGaze	5.19°
			UT-Multiview	4.58°
	Blinking	[57]	unknown	0.59°
		[58]	RT-BENE(RT-GENE with blink labels)	AP:0.997
Insufficient Training Samples	Synthetic Images	[59]	MPIIGaze	4.5°
			UnityEye	4.5°
		[60]	UT-Multiview	8.4°
			MPIIGaze	7.5°
		[61]	unknown	2.49°
		[62]	MPIIGaze	9.95°
		[63]	MPIIGaze	7.9°
		[64]	U2Eyes	~
		Gaze Redirection	~	~
Some Novel Ideas	Landmarks and Semantic Information		MPIIGaze	4.56°
		[65]	ColumbiaGaze	3.59°
			EYEDIAP	10.63°
		[66]	UT-Multiview	5.7°
			EYEDIAP	7.725°
		[67]	ColumbiaGaze	2.22°
	Two-eye Asymmetry	[68]	EYEDIAP	13.5°
			MPIIGaze	8.8°
		[69]	MPIIGaze	4.3°
			EYEDIAP	5.71°
			RT-GENE	8.4°
		[70]	MPIIGaze	4.1°
	Dynamic Information		EYEDIAP	5.3°
		[71]	unknown	92.80%
		[72]	unknown	2.37°
		[73]	unknown	23.82°
		[74]	Gaze360	13.5°
		[75]	MPIIGaze	4.18°
			EYEDIAP	5.84°
		[76]	EYEDIAP	FT: 5.19° SC: 3.4°

of head pose, potential interested objects, gaze directions are modeled temporally by a graph model. The experimental results show that the temporal information is positive for encoding accurate gaze information. Cantariño *et al.* [72] analyzed the improvements of gaze estimation with temporal information using long short-term memory network (LSTM). The results show that temporal information is particularly beneficial to improving accuracy on vertical axis estimates. Krahnstoever *et al.* [73] proposed a joint estimation for body pose and gaze direction in the wild, which need four fixed cameras and four pan tilt zoom (PTZ) cameras. The body position is detected by four fixed cameras, and then four PTZ cameras are controlled to zoom in the subject's face to estimate head pose. Gaze direction is obtained by calculating rotation matrix from HCS to WCS. Finally, sequential Monte Carlo Filtering and Markov Chain Monte Carlo sampling are used to estimate body pose and gaze direction dynamically. Kellnhofer *et al.* [74] proposed an open gaze data set with

temporal information. A baseline is proposed and it achieves the mean error of 13.5° , which use LSTM to dynamically encode gaze information extracted from the backbone ResNet-18. Zhou *et al.* [75] estimated gaze from a static module and a temporal module. The static module learns features from the face and eye appearances while the temporal module, bi-LSTM, encodes temporal gaze information from the output of the static module. The proposed method achieves the mean error of 4.18° and 5.84° on the MPIIGaze and the EYEDIAP. Palmero *et al.* [76] extracted gaze representation from eye regions and full faces using CNN and fed the learned gaze representation into recurrent neural network. The mean error of 5.19° and 3.4° are achieved using this method in the task of FT and CS on the EYEDIAP.

Table III illustrates the methods addressing blinking, occlusion, blur images, insufficient training samples, and other novel ideas from used methods, references, evaluated data sets and the accuracy. For blur images and lighting variations,

TABLE IV
COMPARISON OF 3-D GAZE ESTIMATION DATA SETS

Dataset	Channel	Full Face	Camera Number	Total	Subject	Distance	Gaze Pitch/Yaw	Head Pose Pitch/Yaw
MPIIGaze[15]	RGB	No	1	213,659	15	40-60 cm	P[-48.94°, 13.05°] Y[-52.38°, 44.07°]	P[-51.39°, 62.14°] Y[-73.14°, 73.14°]
MPIIFACEGaze[28]	RGB	Yes	1	213,659	15	40-60 cm	P[-39.25°, 15.18°] Y[-45.26°, 36.99°]	P[-45.58°, 66.32°] Y[-81.71°, 101.93°]
ColumbiaGaze[77]	RGB	Yes	5	5880	56	200 cm	P[-15.00°, 15.00°] Y[-10.00°, 10.00°]	P[-00.00°, 00.00°] Y[-30.00°, 30.00°]
EYEDIAP[78]	RGB-D	Yes	2	94 videos	16	80-120 cm	P[-84.68°, 66.31°] Y[-162.28°, 168.91°]	P[-48.95°, 66.68°] Y[-95.99°, 94.80°]
Gaze360[74]	RGB	Yes	5	≈172K	238	≈ 220 cm	P[-78.22°, 89.26°] Y[-179.88°, 179.88°]	~
NISLGaze[41]	RGB	Yes	1	2079 videos	21	90 cm	P[-21.48°, 20.76°] Y[-21.25°, 21.04°]	~
RT-GENE[55]	RGB-D	Yes	1	277,286	15	≈ 182 cm	P[-80.76°, 34.29°] Y[-50.89°, 66.69°]	P[-63.39°, 26.70°] Y[-37.49°, 37.50°]
ShanghaiTechGaze[56]	RGB	Yes	3	233,796	137	~	P[0.09 cm, 33.44 cm] Y[2.24 cm, 58.36 cm]	~
UnityEye[62]	RGB	No	~	user-defined	~	user-defined	user-defined	user-defined
UT-Multiview[5]	RGB	No	8	1,216,000	50	60 cm	P[-66.48°, 55.43°] Y[-77.92°, 80.80°]	P[-35.51°, 41.93°] Y[-39.90°, 38.78°]
XGaze[79]	RGB	Yes	18	1,083,492	110	100 cm	P[-89.69°, 89.21°] Y[-178.99°, 177.05°]	P[-89.50°, 89.31°] Y[-148.27°, 175.6°]

scale-incentive gaze features can be extracted to mitigate this problem. For occlusions, glasses can be removed using GAN in original images, and self-occlusion can be solved by multiview data. For blinkings, a specific eye blink detection mechanism should be used to remove outlier samples both in training stages and online estimation stages, to ensure gaze representation learned from useful data. Synthetic images can be used as a way of data augmentation and these synthetic data can be generated using Unity tools, perspective transformation, redirection methods and GAN. Novel ideas like using landmarks and semantic information, two-eye asymmetry, and dynamic information can also improve gaze estimation. Gaze can be modeled alternatively from eye landmarks and simplified 2-D eye maps since these features also encode gaze information. The two-eye asymmetry methods weight one of the eyes for gaze estimation since the performance of both eyes is different. Dynamic eye movements encoding temporal context information is also useful for gaze estimation.

III. DATA SETS

This section reviews the existing data sets for 3-D gaze estimation in terms of the distribution of angles and configurations. Comparison of different data sets is shown in Table IV. However, samples with inaccurate annotations would lead to abnormal head poses and gaze angles. Hence, 2-D histogram maps of each data set is further shown in each section to indicate the majority of head pose and gaze.

A. MPIIGaze and MPIIFACEGaze

The MPIIGaze data set contains 213 659 images (60×35 pixels for each eye) of both eyes from 15 participants (six females and five with glasses) with varying illuminations and eye appearances, which is collected using laptops during their daily routine. Participants are asked to fixate on moving dots showed on a screen. The number of images collected by each participant varied from 34 745 to 1498. The

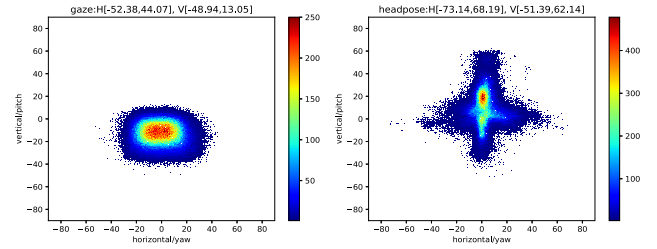


Fig. 5. 2-D distribution of gaze and head pose for the MPIIGaze.

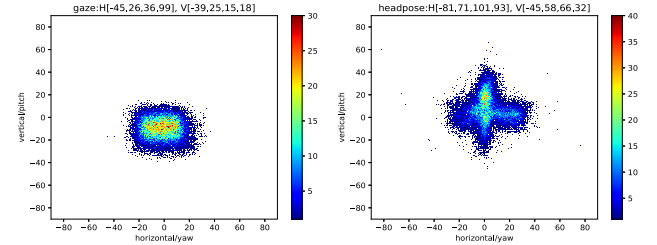


Fig. 6. 2-D distribution of gaze and head pose for the MPIIFACEGaze.

MPIIFACEGaze data set contains full face (1280×720 pixels) images of subjects and other characteristics are the same as the MPIIGaze. Figs. 5 and 6 summarize distributions of head pose angles and gaze angles in the original space.

B. ColumbiaGaze

The ColumbiaGaze data set contains 5880 high-resolution images (5184×3456 pixels) of 56 people (24 females and 21 with glasses). Participants are seated in a fixed location in front of a black background using a chin rest to stabilize their head with 70 cm above the floor. Then, they are asked to gaze at static dots placed in 5° increments horizontally and 10° increments vertically on the blackboard. Thus, the ground-truth gaze directions and head poses tend to be scattered, as shown in Fig. 7. The collected samples are checked to ensure no blinking.

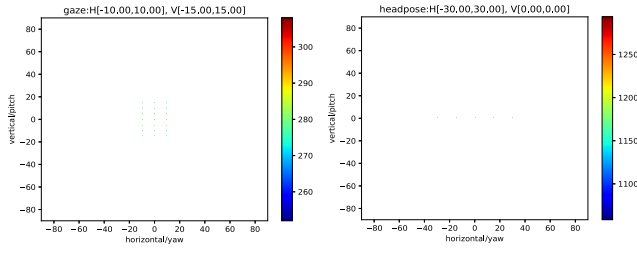


Fig. 7. 2-D distribution of gaze and head pose for the ColumbiaGaze.

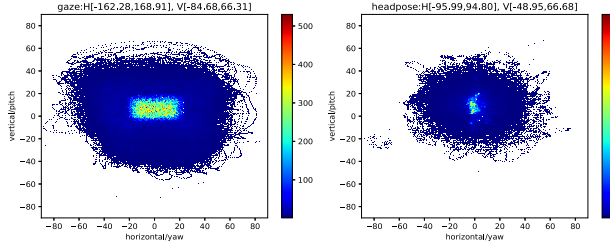


Fig. 8. 2-D distribution of gaze and head pose for the EYEDIAP.

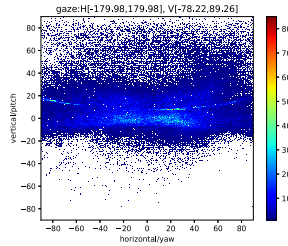


Fig. 9. 2-D distribution of gaze and head pose for the Gaze360°.

C. EYEDIAP

The EYEDIAP data set contains 94 videos of 2–3 min with 640×480 and 1920×1080 pixels from 16 subjects (12 males and four females) using a Kinect sensor and a HD camera. Five LED-markers visible to both cameras are used to synchronize the streams of both cameras. Subjects are asked to gaze at a floating ball and moving targets on a screen with static and head movements, various illuminations and distances to the camera. Fig. 8 summarizes distributions of head pose angles and gaze angles while subjects gaze at a discrete screen target and a 3-D floating target. The distribution of data while subjects gaze at a continuous screen target is not considered here cause the eye state is not provided in this task.

D. Gaze360°

The Gaze360° data set contains around 172 000 images (3382×4096 pixels) collected from 238 subjects under indoor and outdoor conditions. The data set provides full body images and cropped images from face regions. During the data collection under outdoor conditions, subjects are asked to gaze at a small cross which at the bottom of a moving white board with a large AprilTag. The AprilTag is used to find the 3-D location of the cross to further calculate the ground truth. Because cameras record data while moving around subjects, some samples are lack of frontal face regions. Fig. 9 shows the distribution of gaze angles.

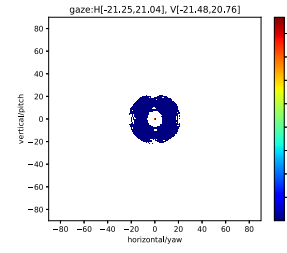


Fig. 10. 2-D distribution of gaze for the NISLGaze.

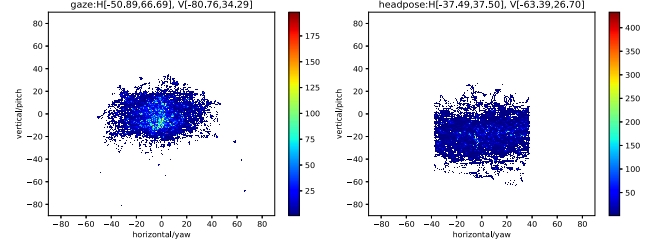


Fig. 11. 2-D distribution of gaze and head pose for the RT-GENE.

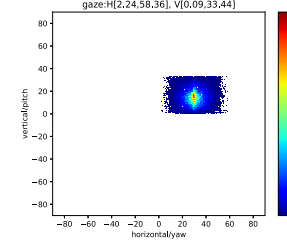


Fig. 12. 2-D distribution of gaze points for ShanghaiTechGaze.

E. NISLGaze

The NISLGaze data set collects 2079 videos from 21 subjects (ten females and ten with glasses) with various gaze directions, head poses and head positions. During the collection, subjects are instructed to gaze at nine gaze blocks sequentially while keeping the body location fixed. Each image is at a resolution of 1920×1080 pixels. Fig. 10 shows the distribution of gaze angles in normalized space II.

F. RT-GENE

The RT-GENE data set contains recordings of 15 participants (nine males and all with glasses), with a total of 122 531 labeled training RGB images (1920×1080 pixels), depth images (512×424 pixels) and 154 755 unlabeled images from the same subjects where the eyetracking glasses are inpainted using GAN. RGB-D videos are collected using a kinect sensor and ground truths are annotated using the eyetracking glass. The camera-subject distances range between 0.5 and 2.9 m. Fig. 11 shows the distribution of gaze angles.

G. ShanghaiTechGaze

The ShanghaiTechGaze record 233 796 images (90×90 pixels) from 137 participants (98 males and 39 females) using three cameras. During the recording, subjects are asked to fixate on moving points on a screen. Fig. 12 shows the distribution of gaze points.

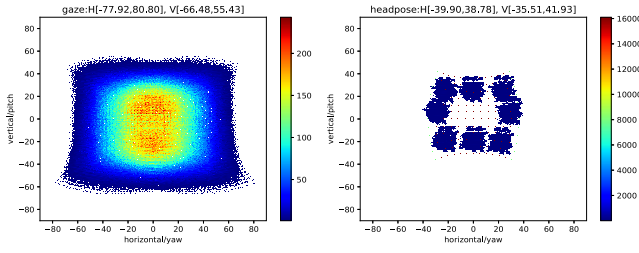


Fig. 13. 2-D distribution of gaze and head pose for the UT-multiview.

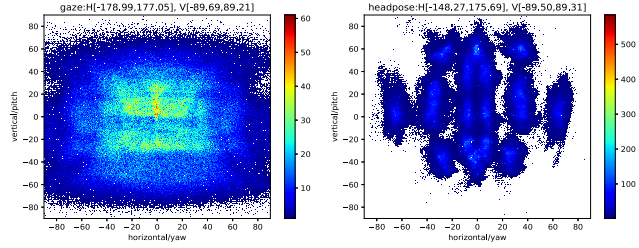


Fig. 14. 2-D distribution of gaze and head pose for XGaze.

H. UnityEye

The UnityEye is a tool for generating synthetic single-eye images, which is based on the Unity software. The resolution of synthesized eye images can be defined by choosing the screen solution when starting the software. Gaze directions and head poses of synthesized eye images can also be defined through eye parameters and camera parameters in this software. Generated samples are shown in Fig. 2.

I. UT-Multiview

The UT-multiview data set contains 1 152 000 synthetic eye images and 64 000 real left and right eye images from 50 subjects (15 females). Eye images are gray-scale with the resolution of 36×60 pixels. A chin rest is used to stabilize the head position located at 60 cm apart from the monitor. Fig. 13 shows the distribution of gaze angles.

J. XGaze

The XGaze is a recently released data set with high resolution RGB images (6000×4000 pixels). 18 HD cameras are used for capturing visual information from different viewpoints. 110 subjects (47 females) are required to watch changing stimulus on a screen with a head rest avoiding unnecessary movements. This data set is with a wide range of gaze and head pose, as shown in Fig. 14.

In summary, characteristics of each data set are shown in Table IV, which compares each data sets from collected channels, whether to have full face, used cameras, total samples, subjects, the distance between subjects and cameras, the range of gaze and head pose. Also, in Figs. 5–14, the 2-D distribution of gaze and head pose for each data set is drawn to show the major distribution. Because the ground truth is calculated using dots shown on a screen, the distributions of gaze and head pose in the MPIIGaze, the MPIIFACEGaze, the NISLGaze, and the ShanghaiTechGaze are limited. In comparison, the distribution of gaze and head pose in the Gaze360° data set is the

most extensive cause the camera capture images while rotates around the subject. A chinrest or head rest is used in the collection process in the UT-multiview, the ColumbiaGaze and the XGaze data set to ensure no unexpected head movements. In contrast, the ground truth of eye gaze is collected using a pair of eyetracking glasses in the RT-GENE, and in the EYEDIAP using a floating ball, which allows for natural head movements. Furthermore, the UnityEye can generate synthetic eye images in a short time when data is limited.

IV. GAZE ESTIMATION APPLICATIONS

Gaze estimation can be applied to HCI, drivers' distraction detection, analysis of human social behaviors and video games.

Valenti *et al.* [31] presented an unconstrained visual gaze estimation system which combine head pose and the displacement of pupil center to capture human visual field. Based on this research, a vision-based multimodal HCI system which exploit hand gesture, eye gaze, head pose and mouth motions is proposed in [80]. Using this system, two applications, a 3-D file system navigator and a geographical information visualization application, are developed. Gaze-based eye-typing [81]–[83] is a novel HCI technology, which can be helpful for disabled people and wearable computing. Gaze estimation can also be applied to monitor children's behavior [84], analyze human's intention [85] and interact with ASD children [71], [86], [87]. Gaze as a nonverbal interaction channel can also be used in video games [88], [89] and human-robot interaction [90], [91]. Additionally, drivers' attention and distraction can also be detected by gaze to ensure safe driving [92]–[97].

V. FUTURE DIRECTIONS TOWARD SOCIAL INTERACTION

A. Metalearning

Metalearning is a science of training model, which is a hot topic in machine learning during past years. In computer vision, metalearning has been successfully applied in image classification [98], object segmentation [99], and detection [100]. It is well known as learning to learn, which can learn prior knowledge and generalize well on new samples. It is hard to give a unified formulation of metalearning, but it can be understood by compare it with conventional machine learning. Conventional machine learning approaches assume that training data and testing data share the same distribution, hence given abundant training samples, conventional algorithms like CNN can learn a mapping function with optimal parameters fitting the training data. However, metalearning aims to learn a set of parameters which can generalize well on unseen samples. The conventional approaches focus on improving the performance of the model on training data, while metalearning pays more attention to the potential of the model in testing data. Therefore, metalearning can be used as transfer learning, few shot learning [101], [102] and reinforcement learning [103] when there are no sufficient samples.

Deep-learning-based gaze estimation also face the issue of insufficient training samples. The collection of gaze samples is time-consuming. Besides, in gaze estimation, person-specific offsets exist in eyeballs which is hard to be learned from

RGB images. Calibration samples are usually used to fine-tune the trained model to eliminate the person-dependent offsets. However, pretrained models tend to be overfitting when limited calibration samples are available. Metalearning is capable of learning a set of optimal weights for the model, which allows fine tuning the model with only few new samples. Few studies [47], [104] investigate the potential of metalearning on gaze estimation. Metalearning for gaze estimation should be considered more in the future.

B. Causal Inference

Pearl [105] indicates that conventional statistics-based machine learning methods can not reason. In his theory of the causal hierarchy consisting of association, intervention and counterfactuals, statistics-based machine learning methods are in the first level association, because they process the raw input data without encoding causal relationships. For example, when training data and testing data are not independent and identically distributed, which is the agnostic data selection bias, correlation-based machine learning approaches propose transfer learning, domain adaptation and metalearning but still can not reason. However, causality-based causal inference can learn a model with causal variables which are insensitive to the selection bias.

Causal inference has been applied to computer vision over the past years, like visual question answering [106]–[110], classification [111], [112], and visual cognition [113]. Gaze estimation face the non i.i.d hypothesis especially in the wild and the gaze-based cognition process also need to be reasoned and inferred from gaze behaviors. Causal reasoning can be applied in social behavior reasoning based on gaze in future.

C. Disentangled Representation

Representation learned in abstract level from deep neural networks is with high descriptive ability but with low interpretability. This kind of representation contains multiple contents like textures and shapes. However, a model or representation trained in this fashion can be easily confused when the model classifies the same objects having the same shape but with different textures [114]. A better solution is to learn or disentangle the shape and texture, the independent variables, from the representation of appearances like [115], disentangling the pose and appearance from the original space.

According to [116], based on the group theory and symmetry transformation, disentangled representation is a vector representation which can be decomposed to independent subspaces and are unaffected by each other, such as poses and appearances [47], [115], [117]. Existing studies have applied this methodology to their field, such as face recognition [117], [118], gait recognition [115], [119], event detection [120], and gaze estimation [47]. Since true gaze can be disentangled into head pose and gaze, disentangled representation may be a solution for gaze estimation under large head poses in future.

D. Social Gaze Behavior

The gaze direction extracted using above mentioned estimation methods could be used to infer gaze behaviors in social interaction scenes. Among all nonverbal information such as pose, gesture, emotion etc, gaze behaviors play an essential role during face-to-face interaction and communication, as gaze reflects the underlying process of mental cognition. For example, strangers always make eye contact with pedestrians before asking for directions, showing their will to start a conversation.

Several researchers have defined and updated the taxonomy of social gaze interaction behaviors [121]–[123]. To better understanding it, a definition of social gaze behavior with hierarchy is specifically illustrated here. According to [122], social gaze interaction behaviors can be grouped into atomic-level behaviors and event-level behaviors. Atomic-level behaviors contain six dyadic gaze states, which are *Single*, *Mutual*, *Avert*, *Refer*, *Follow*, and *Share*. By combining these states according to a certain sequential order, event-level behaviors are defined as *Noncommunicative*, *Mutual Gaze*, *Gaze Aversion*, *Gaze Following*, and *Joint Attention*. This taxonomy indicates a more specific direction to investigate the relationship between gaze and social behaviors using computer vision.

Besides the study on gaze communication [122], other researches study social gaze using the vision technology for specific applications. Massé *et al.* [124] investigated the relationship between gaze, head pose and the visual focus of attention using Bayesian dynamic formulation in social interaction in third-person view. One of the major action of gaze behavior is mutual gaze or eye contact, Chong *et al.* [86] used eyeglasses in natural interactions to capture children's face and detect eye contact to assess the behavior of children with ASD, while Qiu *et al.* [125] designed a eye-tacking glass to simulate gaze contact for blind people for better communication experience. In social scenes, multiple cameras are always used. Chen *et al.* [85] analyzed rapport or hostility of subjects using nonverbal cues like pose, motion and gaze in a multiview system. Capozzi *et al.* [126] investigated the relationship between gaze behaviors and leadership in multiparty configurations. Social gaze can also be alternatively inferred from poses and appearances of head instead of 3-D gaze estimation using eye appearances, such as [127] and [128]. Social gaze is a window toward mental cognition, which can further be studied with neural mechanics. A review about gaze and neural mechanics can be found in [129]. Most 3-D gaze estimation methods do not take social cues into consideration. Therefore, to analyze social gaze behaviors based on 3-D gaze estimation, gaze estimation in unconstrained social interaction scenes should be considered in future.

VI. CONCLUSION

Vision-based gaze estimation is attractive for researchers in computer vision and HCI increasingly. This review analyzes primary problems in gaze estimation, including head pose variance, person-specific bias, eyeblink, occlusion as well as blur issues and summarizes existing prevailing solutions to these problems. Besides, details of public data sets for 3-D

gaze estimation and gaze applications are surveyed. Moreover, we predict advanced algorithms in machine learning which would be exploited in future for social gaze estimation and interaction.

Head movements change eye appearances and HCS movements degrade estimation accuracy. The bias between optical axis and visual axis is person dependent and is hard to be estimated using RGB images solely, thus calibration is required. Occlusion, blink, and blur issues also decrease the generalization ability of pretrained models. In social gaze interaction, the influence of these problems is more obvious, since social gaze is estimated without any constraints. Therefore, few-shot learning, casual inference and disentangled representation are introduced here as possible solutions to infer gaze direction and behaviors in social interaction. Our future works will focus on 3-D gaze estimation in unconstrained social interaction and gaze-based social behavior reasoning.

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